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ARIN330585 – Trí tuệ nhân tạo

Introduction to Artificial Intelligence

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Chapter 3: Ethics and Law

1. *AI Ethics*
2. *AI Law*

Bad and Good

BAD	GOOD
× Killing	✓ Saving lives
× Stealing	✓ Helping others in need
× Being lazy, drunk	✓ Learning, working hard
× Lying	✓ Being honest
× Having an extramarital relationship	✓ Being faithful

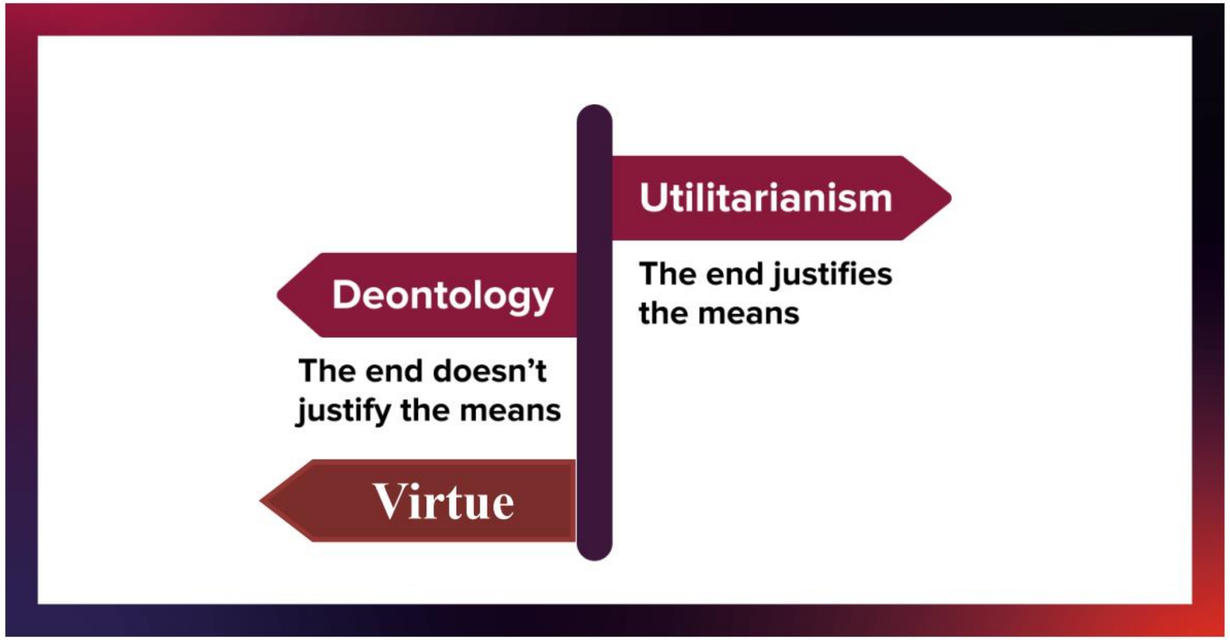
- Ethics, also called moral philosophy, is a branch of philosophy that studies right and wrong behavior. It talks about what is good and bad, what is just and unjust, and so on.

- In short, the goal of ethics is to propose values that we can pursue to live a life free from regrets.

Basic principles of AI normative ethics

Normative ethics regulate behavior. It tell us what the right or the wrong thing to do is. In other words, it provides a standard, rule, or criterion for judging right and wrong. The Golden Rule is an example of normative ethics.

AI ethics is an actively researched field today. Among the many ethical frameworks, three major approaches are virtue ethics, deontology, and utilitarianism.

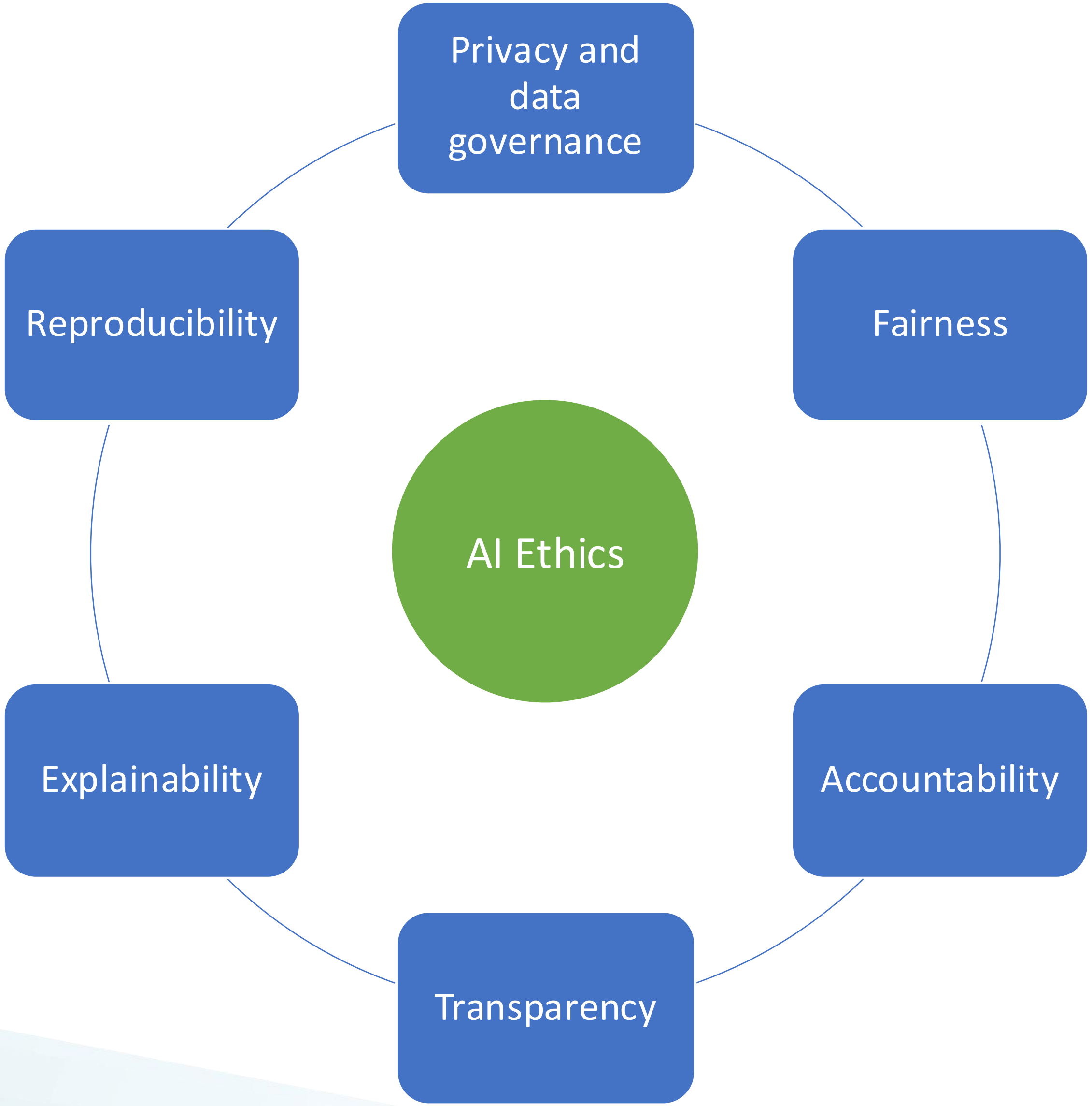


Utilitarianism: chủ nghĩa vị lợi

Deontology: nghĩa vụ/bổn phận

Virtue: đức hạnh

Ethical principles of AI



In AI ethics, responsibility is understood in three dimensions:

- Which people (or groups) are accountable for the impact of AI?
- How can society use and develop AI in a way that is legally and morally acceptable?
- What moral and legal responsibility should we demend from the AI system itself?

Data analytics and Law

United States: privacy laws (e.g.,HIPAA 1996, COPPA 1998, FACTA 2003), Negligence laws

European Union: General Data Protection Regulation (GDPR), some main rules:

- Data analytics must be fair.
- You need permission to process data.
- Individuals are allowed to access the personal data and correct if it's not correct.
- Accountability.
- ...

reference: Ethics and Law in Data and Analytics, edx.org/course/ethics-and-law-in-data-and-analytics-2
https://www.defense.gov/News/Releases/Release/Article/2091996/dod-adopts-ethical-principles-for-artificial-intelligence/

IRAC

A traditional method for analyzing legal problems includes 4 parts: **Issue, Rule, Application,** and **Conclusion** (IRAC)

- Issue: define the **legal problem**
- Rule: identify **laws that are applicable** to the Issue
- Apply: **analyze the Issue** according to the Rule
- Conclusion: make **legal conclusion** to the Issue

Main reference: Ethics and Law in Data and Analytics, by Microsoft, edx.org/course/ethics-and-law-in-data-and-analytics-2

How to build responsible AI

Values & Risk → Data Quality → Fair & Transparent Design → Human Oversight → Testing & Monitoring → Accountability → Documentation

Don't start with the question: is the model highly accurate?

Instead, add:

- Who will be affected?
- What are the potential risks?
- If it's wrong, who will be harmed?

How to build responsible AI

Values & Risk → **Data Quality** → Fair & Transparent Design → Human Oversight → Testing & Monitoring → Accountability → Documentation

- Check for data discrepancies
- Check if any groups are missing
- Check if labels are reliable
- Check if the data is legitimate and appropriate for its purpose

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- Fairness by design
- Privacy by design
- Transparency by design

How to build responsible AI

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- AI provides support, but it doesn't always replace humans.
- There must be people to supervise, intervene, correct errors, and handle complaints.

How to build responsible AI

Values & Risk → Data Quality → Fair & Transparent Design → Human Oversight → **Testing & Monitoring** → Accountability → Documentation

- Test before deployment
- Audit fairness and safety
- Monitor drift and errors after deployment
- Install an incident reporting mechanism

How to build responsible AI

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- Who approves the system?
- Who is responsible if the system causes harm?
- Who handles complaints?
- Who decides to stop the system?

How to build responsible AI

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- What AI does well?
- What it shouldn't be used for?
- In what context is accuracy important?
- Limitations, biases, operating conditions

Trung tâm Học liệu và Dạy học số

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